

Understanding Odds and Probability

A Simple Illustration Using Credit Card Default

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Problem Statement

This problem examines the relationship between **odds** and **probability** in the context of credit card default.

- (a) On average, what fraction of people with odds of 0.37 of defaulting will actually default?
- (b) If an individual has a 16% probability of defaulting, what are the odds that she will default?

- **Default:** Failure to make a required credit card payment.
- **Probability (p):** Measures how likely default is to occur.

$$0 \leq p \leq 1$$

- **Odds (O):** Compares the chance of default to the chance of no default.

$$O = \frac{p}{1 - p}$$

where $0 \leq O < \infty$

Fraction Defaulting: The proportion of people who default (numerically equal to probability).

Difference Between Probability and Odds:

- Probability answers: *"How often does default occur?"*
- Odds answer: *"How likely is default compared to no default?"*

Key Formulas

The relationship between probability p and odds O is given by:

$$p = \frac{O}{1 + O}$$

$$O = \frac{p}{1 - p}$$

These formulas allow us to convert between odds and probability.

Part (a): Converting Odds to Probability

Given:

$$O = 0.37$$

Step 1: Apply the formula

$$p = \frac{O}{1 + O}$$

$$p = \frac{0.37}{1.37}$$

Step 2: Compute the value

$$p \approx 0.2701$$

Interpretation: Approximately 27.01% of people with odds of 0.37 will default.

Answer to Part (a)

$$p \approx 0.27 \text{ (27.01\%)}$$

On average, about **27%** of individuals with odds of 0.37 will default.

Part (b): Converting Probability to Odds

Given:

$$p = 16\% = 0.16$$

Step 1: Apply the formula

$$O = \frac{p}{1 - p}$$

$$O = \frac{0.16}{0.84}$$

Step 2: Compute the value

$$O \approx 0.1905$$

Answer to Part (b)

$$O \approx 0.19$$

The odds that the individual will default are approximately **0.19**.

- Odds of 0.37 correspond to a default probability of approximately 27%.
- A 16% probability of default corresponds to odds of approximately 0.19.

Conclusion: Odds and probabilities describe the same risk using different mathematical scales.

James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An Introduction to Statistical Learning: With Applications in R* (2nd ed.). Springer.

See Chapter 4: *Classification*, Section 4.3.1: *The Logistic Model*, Equation (4.3).

Thank You